



Certificate of Achievement

Joshuva Jeemon

has completed the following course:

MACHINE LEARNING FOR IMAGE DATA
THE UNIVERSITY OF NOTTINGHAM AND UNIVERSITY OF LINCOLN

This course looked at the use of machine learning using image data, and discussed feature extraction, image labelling and pre-processing. We covered both supervised and unsupervised methods, the common tasks of classification, clustering, and regression, and introduced some deep learning concepts.

5 weeks, 3 hours per week



Andrew French
Professor of Computer Science
The University of Nottingham



**University of
Nottingham**
UK | CHINA | MALAYSIA



The person named on this certificate has completed the activities in the attached transcript. For more information about Certificates of Achievement and the effort required to become eligible, visit futurelearn.com/proof-of-learning/certificate-of-achievement.

This certificate represents proof of learning. It is not a formal qualification, degree, or part of a degree.



Joshuva Jeemon

has completed the following course:

MACHINE LEARNING FOR IMAGE DATA

THE UNIVERSITY OF NOTTINGHAM AND UNIVERSITY OF LINCOLN

The course looked at the use of machine learning, particularly in the context of plant image data collected experimentally, and discussed feature extraction, labelling, and pre-processing. We covered both supervised and unsupervised methods, the common tasks of classification, clustering, and regression, and ways to train, visualise, and evaluate models using the Python packages Scikit-Learn, Matplotlib and Pandas. Finally, we discussed how deep learning techniques differ from machine learning.

STUDY REQUIREMENT

5 weeks, 3 hours per week

LEARNING OUTCOMES

- Explain what machine learning is, and how it relates to image data
- Perform simple image labelling and pre-processing tasks
- Classify supervised, unsupervised and semi-supervised machine learning techniques
- Code some simple machine learning scripts using Python Scikit-Learn
- Describe some common machine learning tasks, such as clustering, regression, and classification
- Investigate deep learning, and how it differs from machine learning

SYLLABUS

- What is machine learning? Key concepts
- Supervised versus unsupervised learning
- Software tools – Python Scikit-Learn
- Common tasks and terminology

- Data and features
- Feature extraction
- Labelling images
- Image pre-processing
- Common techniques
- Clustering
- Classification
- Regression
- Evaluation, visualisation, and selection
- Tips and tricks
- Good training practice
- Data augmentation
- Common challenges
- Introduction to Deep learning
- Deep learning vs machine learning
- Neural networks
- Some simple tools